Program 4: Practical Exercise: Build and Run a Java Application with Maven**,** Migrate the Same Application to Gradle

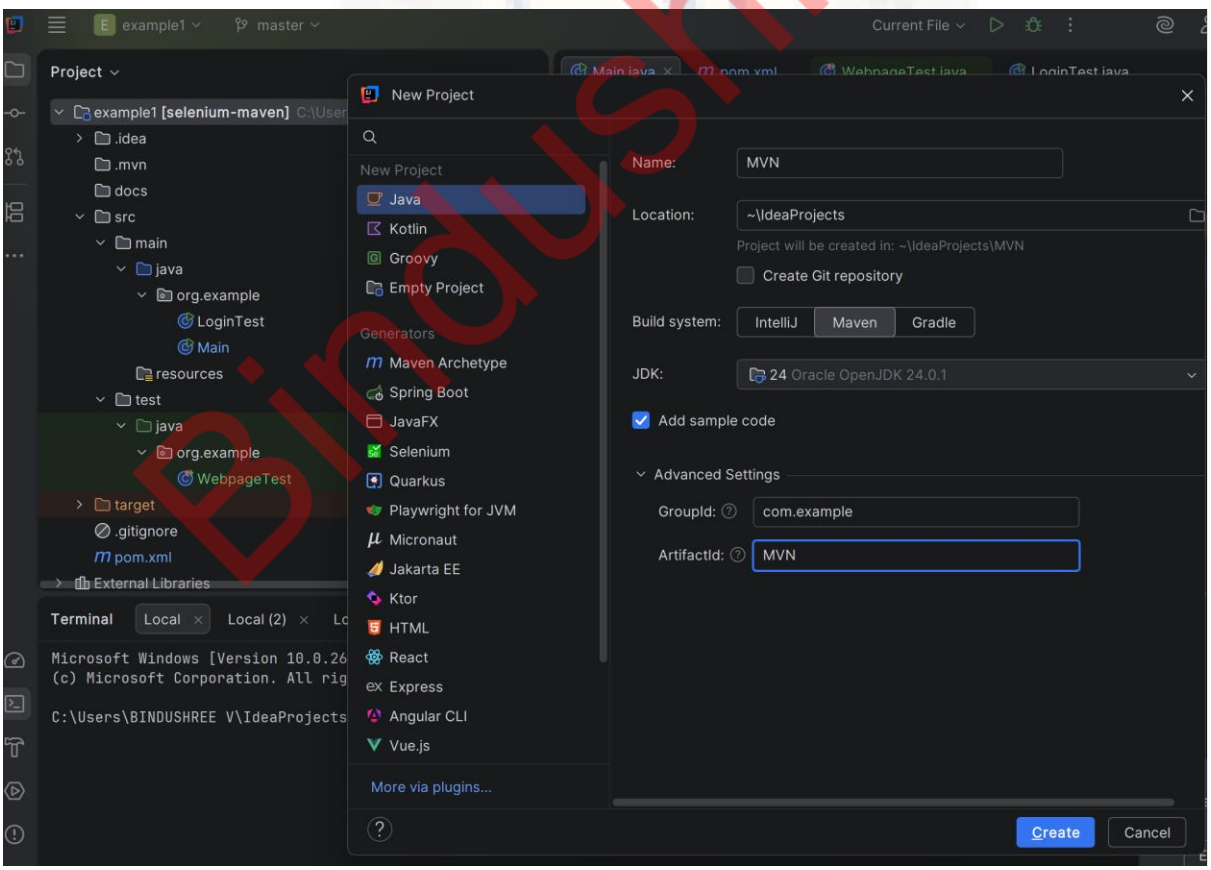
Prerequisites

- JDK 8 or above installed
- IntelliJ IDEA installed
- Maven & Gradle configured (or use IntelliJ's bundled versions)

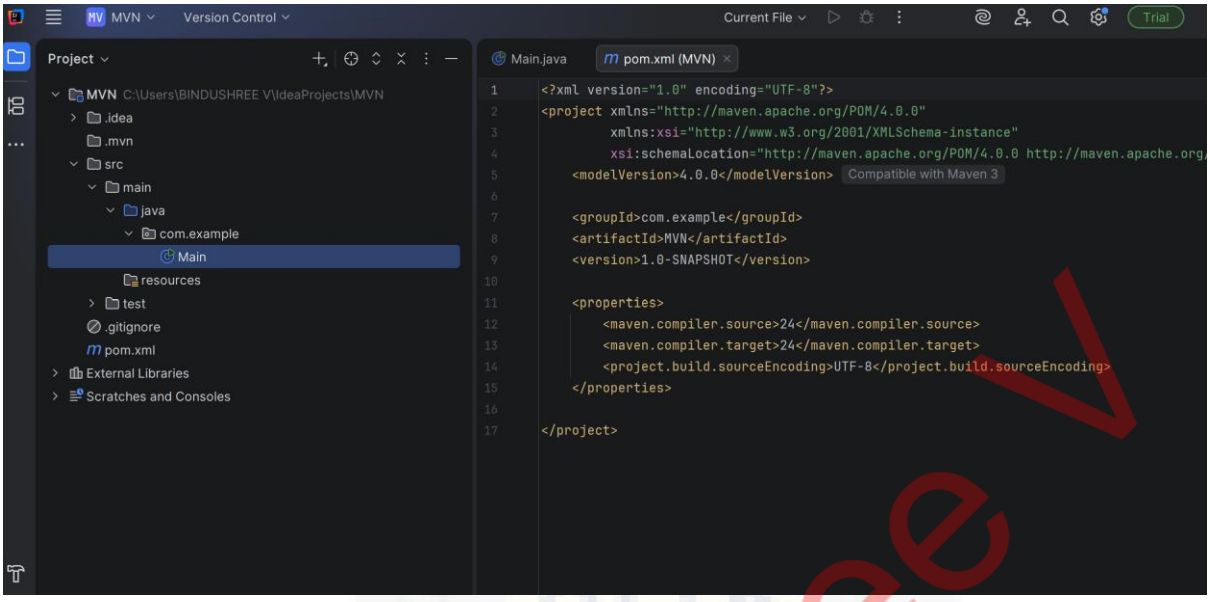
TASK 1: Build & Run Java Application using Maven

Step 1: Create Maven Project in IntelliJ IDEA

1. Open IntelliJ IDEA
2. Click New Project
3. Select Maven
4. Choose JDK
5. Click Next
6. Enter:
   - GroupId: com.example
   - ArtifactId: MVN (or any 3-letter name)
7. Finish



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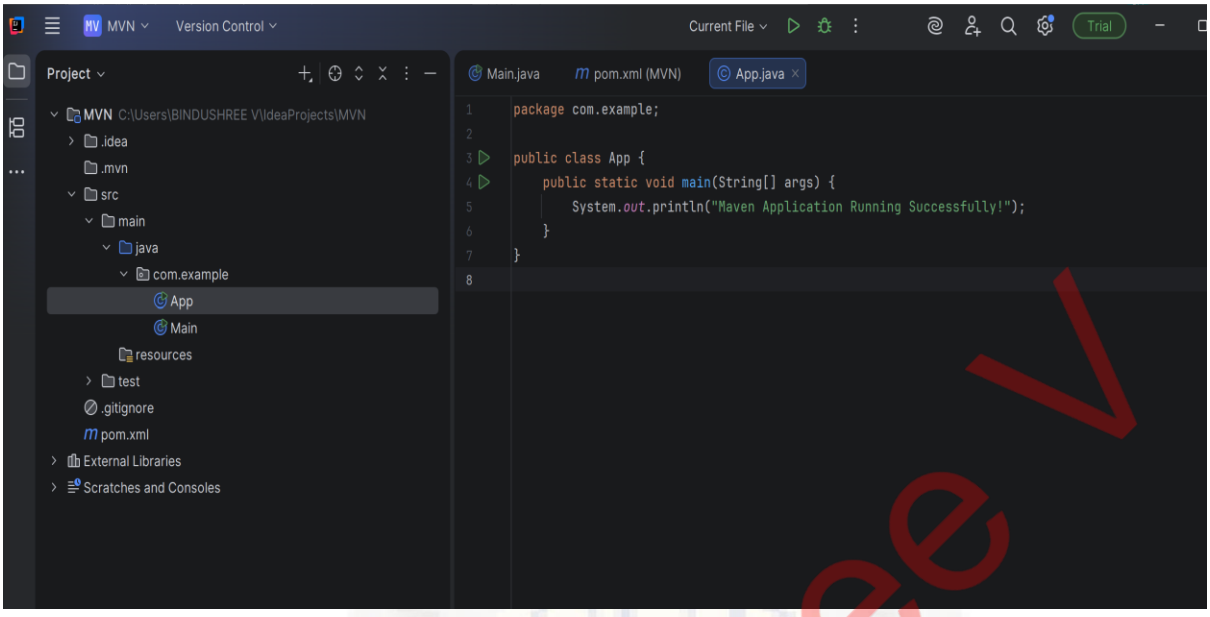
Created MVN Project

Step 2: Create Main Java Class
Create file: src/main/java/com/example/App.java

```
package com.example;

public class App {
    public static void main(String[] args) {
        System.out.println("Maven Application Running Successfully!");
    }
}
```

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Step 3: Add Build Plugins in pom.xml
Paste this code AFTER <dependencies> tag

```
<build>
   <plugins>

      <!-- Compiler Plugin -->
      <plugin>
         <groupId>org.apache.maven.plugins</groupId>
         <artifactId>maven-compiler-plugin</artifactId>
         <version>3.8.1</version>
         <configuration>
            <source>1.8</source>
            <target>1.8</target>
         </configuration>
      </plugin>

      <!-- Jar Plugin -->
      <plugin>
         <groupId>org.apache.maven.plugins</groupId>
         <artifactId>maven-jar-plugin</artifactId>
         <version>3.2.0</version>
         <configuration>
            <archive>
```

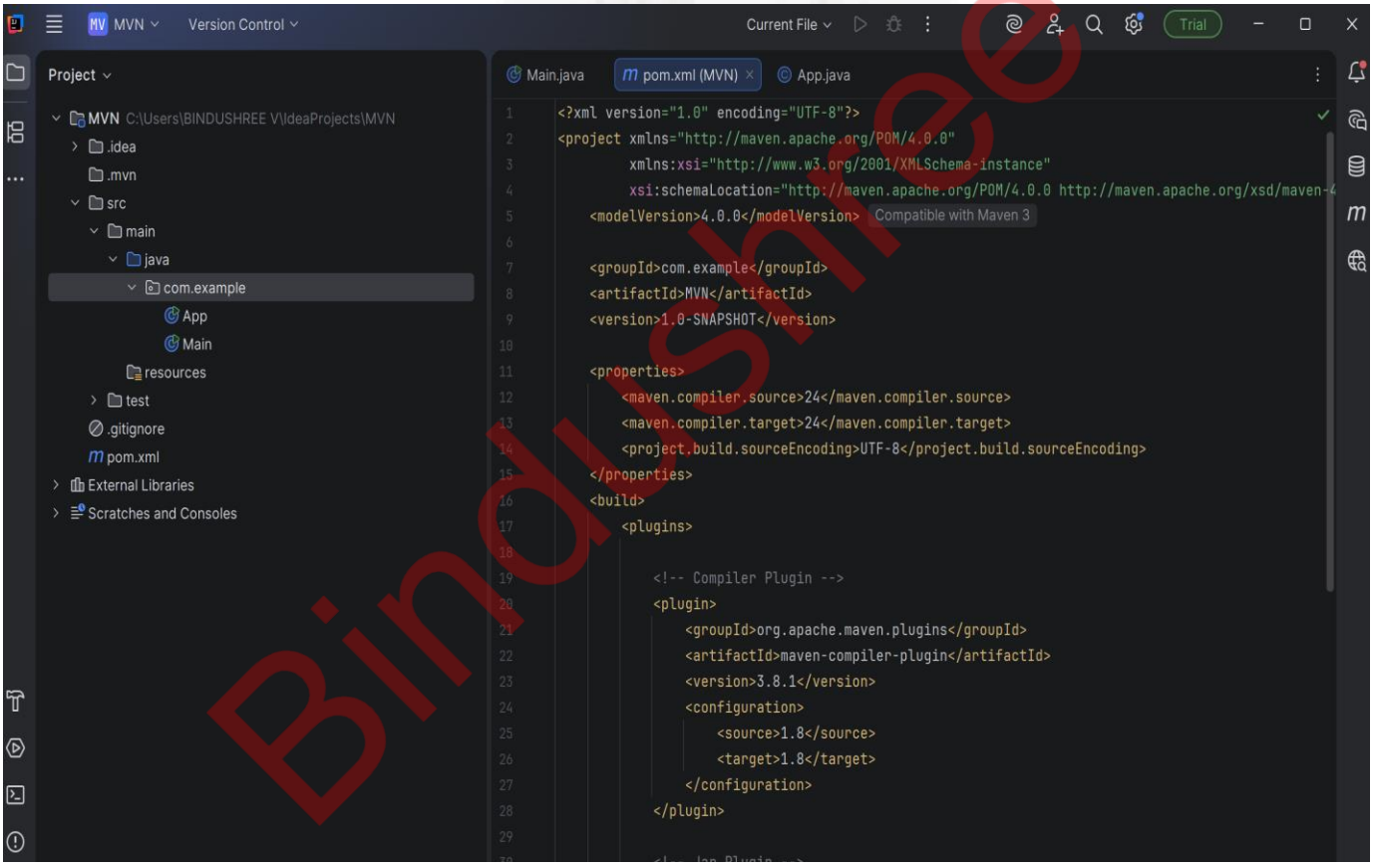
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```
            <manifest>
                <mainClass>com.example.App</mainClass>
            </manifest>
        </archive>
    </configuration>
</plugin>

</plugins>
</build>
```

Why this is needed

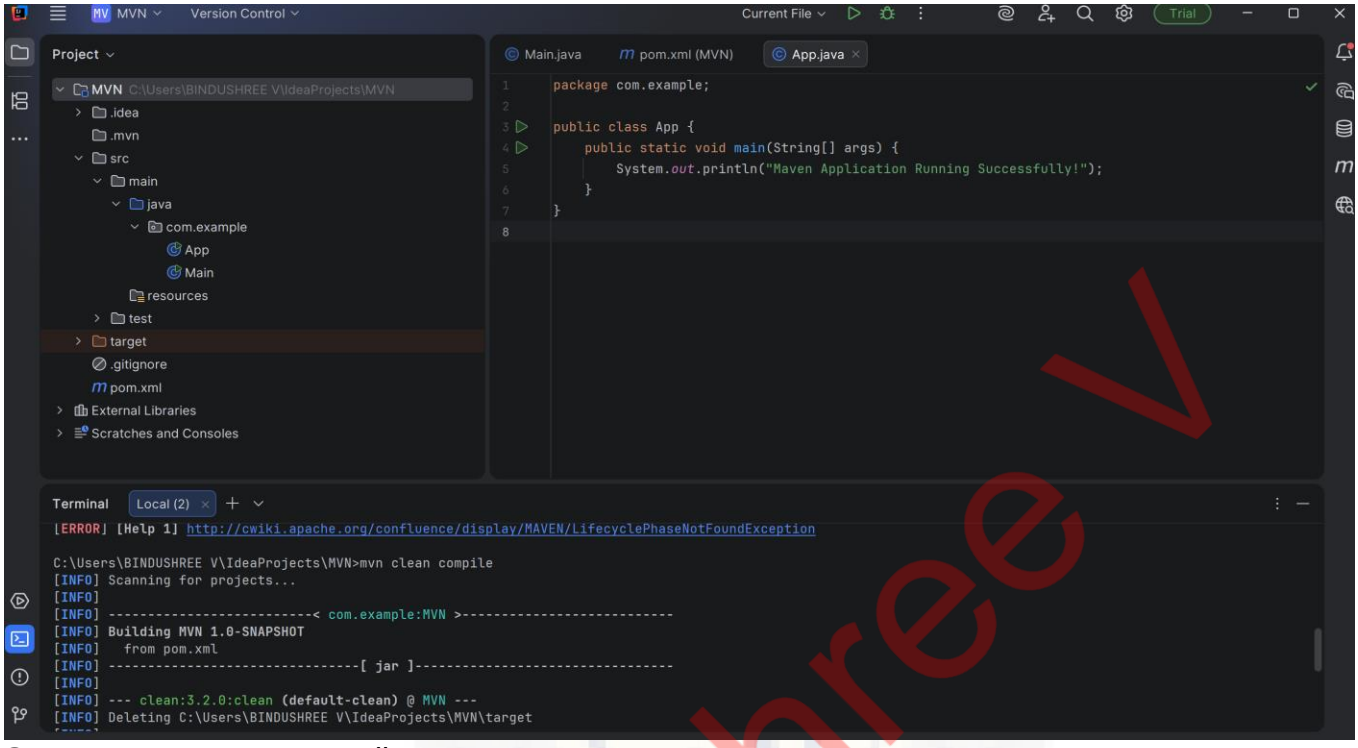
- maven-compiler-plugin → compiles Java code
- maven-jar-plugin → creates executable JAR
- mainClass → tells JVM where execution starts



After adding the plugin reload the maven

Step 4: Build Maven Project
Open terminal in Intellij Idea and run: mvn clean compile, mvn package
JAR file created inside: target/

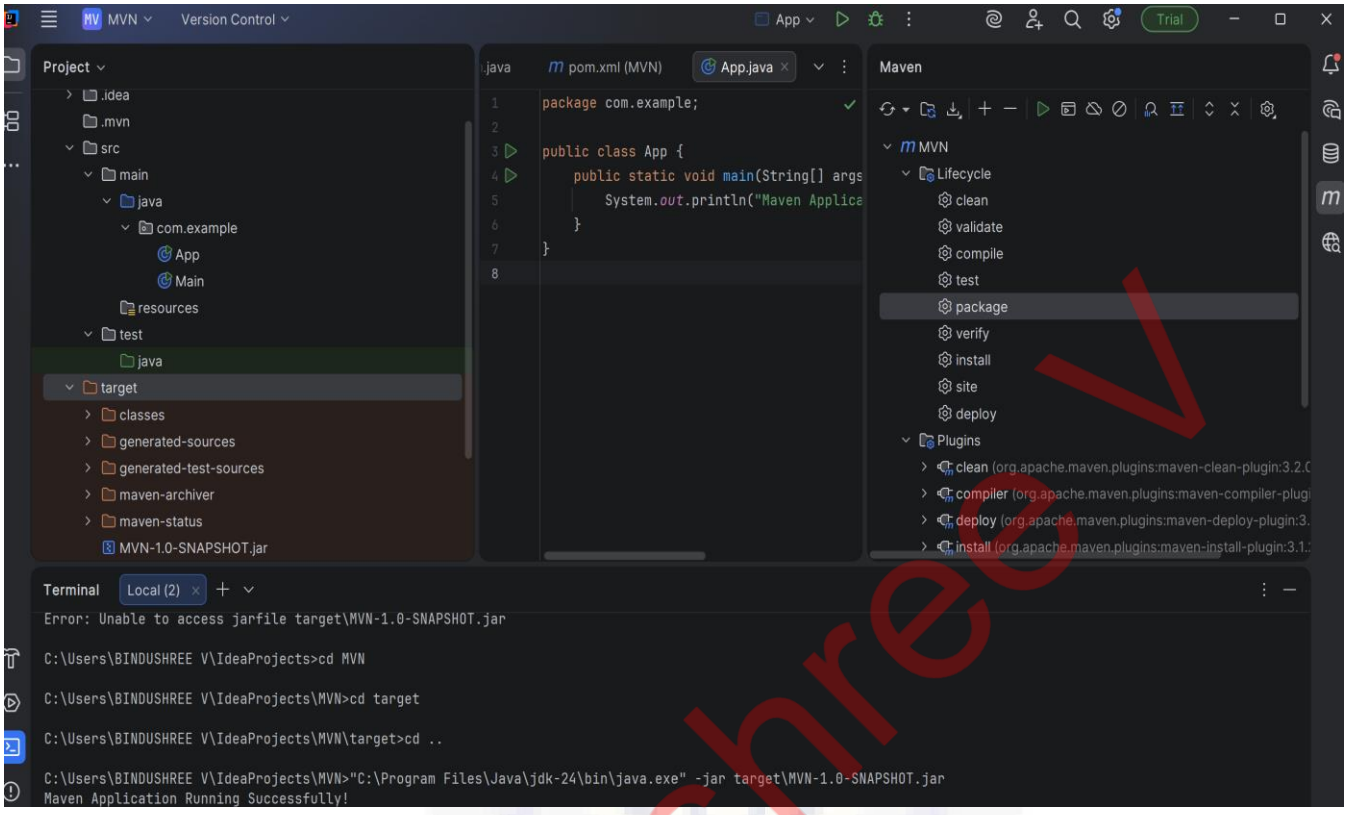
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Step 5: Run Maven JAR File
Go to File → Project Structure → Project
Go to File → Settings → Build, Execution, Deployment → Build Tools → Maven

In the Intellij Idea teminal run : java -jar target/MVN-1.0-SNAPSHOT.jar
if you don't get output by running the above command, run the following
"C:\Program Files\Java\jdk-24\bin\java.exe" -jar target\MVN-1.0-SNAPSHOT.jar

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TASK 2: Migrate Maven Project → Gradle

Step 6: Convert Maven to Gradle

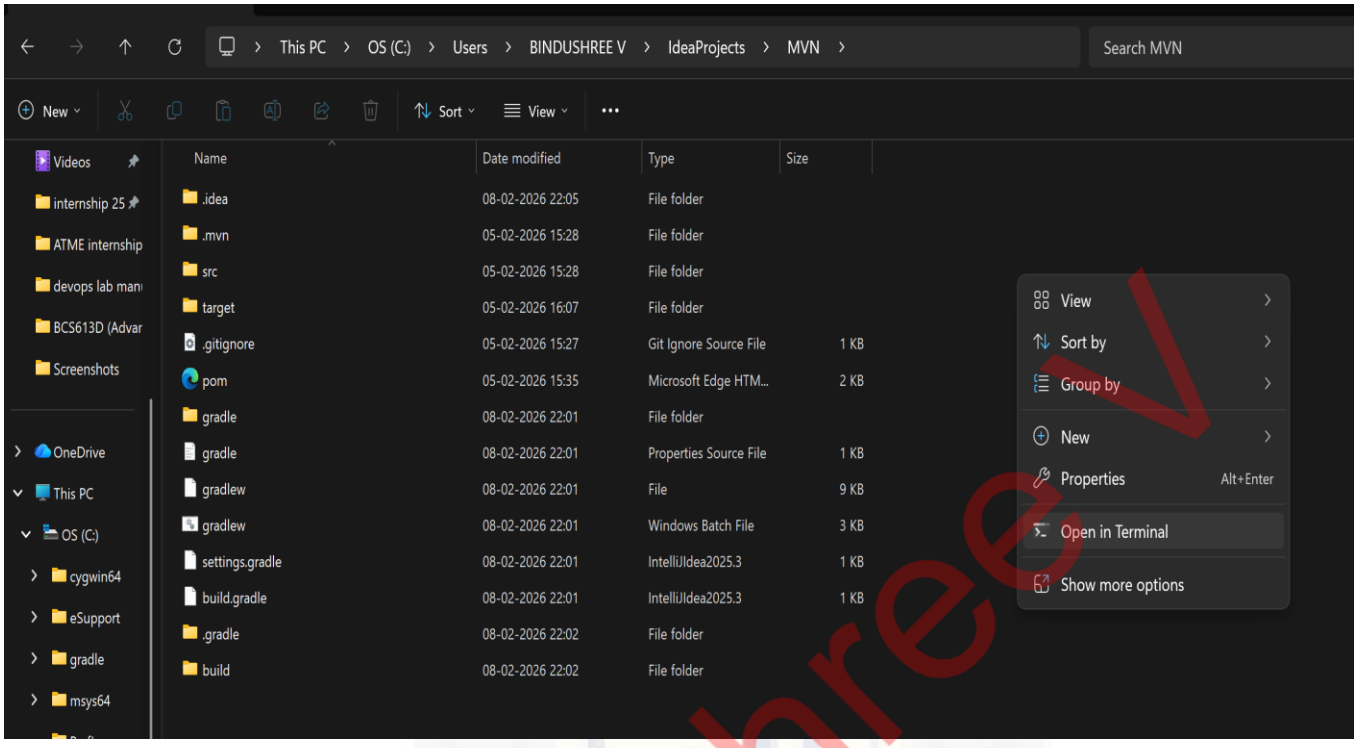
In the **same project directory**, run:

✓ This command:

- Reads pom.xml
- Automatically creates build.gradle
- Migrates dependencies and project structure

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Go to the directory open in terminal and Run: gradle init --type pom



Select Groovy→yes and execute gradle build

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```
PS C:\Users\BINDUSHREE V\IdeaProjects\MVN> gradle init --type pom
Starting a Gradle Daemon (subsequent builds will be faster)

Select build script DSL:
  1: Kotlin
  2: Groovy
Enter selection (default: Kotlin) [1..2] 2

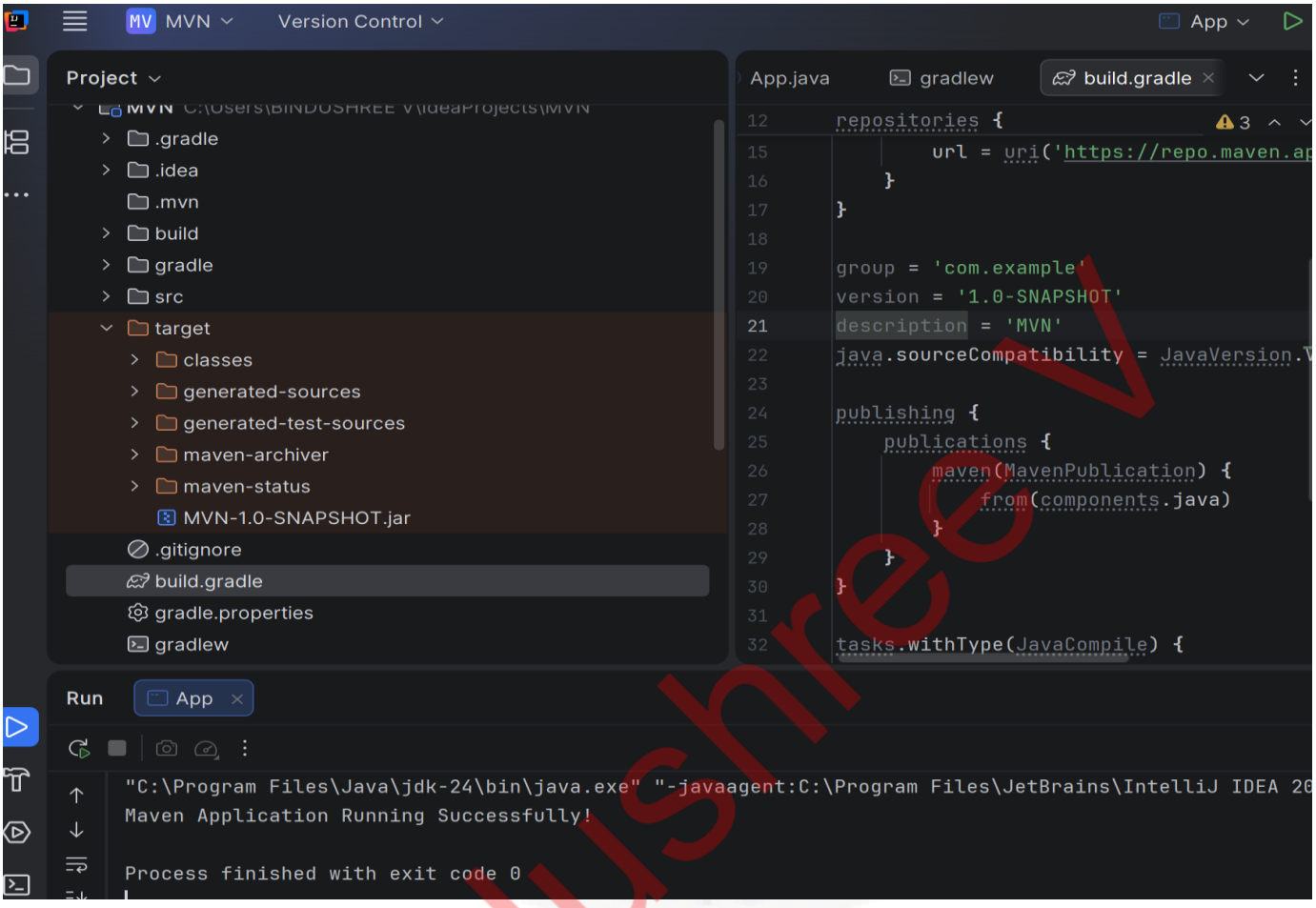
Generate build using new APIs and behavior (some features may change in the next minor release)? (default: no) [yes, no]
 yes


> Task :init
Maven to Gradle conversion is an incubating feature.
For more information, please refer to https://docs.gradle.org/8.14.4/userguide/migrating_from_maven.html in the Gradle d
ocumentation.

BUILD SUCCESSFUL in 42s
1 actionable task: 1 executed
PS C:\Users\BINDUSHREE V\IdeaProjects\MVN> gradle build
Calculating task graph as no cached configuration is available for tasks: build
```

Now go to the IntellijIdea, in your folder you can see build gradle-> run

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